Bright learn
SQL Fundamentals

1 Question
Retrieve all employees and their assigned project using an ~~AA~~ INNER JOIN
- Return : Employee_ID, First Name, Last Name, Department, Salary, projectID, project Name, Budget, Status

```
Select  Employee ID,
        First Name ,
        Last Name ,
        Department,
        Salary ,
        Project ID,
        Project Name,
        Budget ,
        Status
FROM database.schema.employees AS A
Inner JOIN database.schema.~~employees~~ projects AS B
ON A.Employee ID = B.Employee ID
```

Employee ID	First Name	Last Name	Department	Salary	Project ID	Project Name	Budget	Status
1	John	Doe	IT	70 000	101	AI Development	100000	Complete
2	Alice	Smith	HR	60 000	102	Employee Training	50 000	Ongoing
3	Bob	Johnson	Finance	75 000	104	Financial Analysis	90 000	Ongoing
5	Emma	Wilson	Sales	65 000	105	Market expa	65 000	Complete
10	John	Doe	IT	70 000	103	Cyber Security Add	75000	Pending
6	Machael	Clark	Finance	80 000	106	Risk Manager	80 000	Pending

2 Question

Retrieve all employees and ther assigned project, including employees who have no projects using a LEFT JOIN

- return : Employee ID, First Name, Last Name, Department, Salary, Project ID, Project Name, Budget, status

```
Select  Employee ID,
        First Name,
        Last Name,
        Department,
        Salary,
        Project ID,
        Project Name,
        Budget,
        Status
FROM data.schema.employees AS A
LEFT JOIN database.schema.emplproject AS B
ON A.Employee ID = B.Employee ID
```

Employee ID	First Name	Last Name	Department	Salary	Project ID	Project Name	Budget	Status
1	John	Doe	IT	70 000	101	AI Development	100000	Complete
2	Alice	Smith	HR	60 000	102	Employee Training	50 000	Ongoing
3	Bob	Johnson	Finance	75 000	104	Financial Analysis	90 000	Ongoing
4	David	Brown	IT	72 000	null	null	null	null
5	Emma	Wilson	Sales	65 000	105	market expansion	65000	Complete
6	Michael	Clark	Finance	80 000	106	risk management	80000	Pending
1	John	Doe	IT	70 000	103	Cybersecurity	90 000	Ongoing

3 Question

Retrieve all projects and their assigned employees, including projects that have no employees using a RIGHT JOIN

- Return: Project ID, Project Name, Budget, status, Employee ID, First name, Last Name, Department, Salary

```
Select Project ID,
       Project Name,
       Budget,
       Status,
       Employee ID,
       First Name,
       Last Name,
       Salary
FROM database.schema.projects AS A
RIGHT JOIN database.schema.employees AS B
ON A.Employee ID = B.Employee ID
```

Project ID	Project Name	Budget	Status	Employee ID	First Name	Last Name	Department	Salary
101	AI Development	100 000	Complete	1	John	Doe	IT	70 000
102	Employee Training	50 000	Ongoing	2	Alice	Smith	HR	60 000
103	Cybersecurity	75 000	Pending	1	John	Doe	IT	70000
104	Financial Analysis	90 000	Ongoing	3	Bob	Johnson	Finance	75 000
105	Market Expansion	65 000	Complete	5	Emma	Wilson	Sales	65 000
106	Risk Management	80 000	Pending	6	Michael	Clark	Finance	80 000

4 Question

Retrieve all employees and projects, including those without a match in either table using a FULL OUTER JOIN

- Return : employee ID, First name, last name, department, salary, project ID, project name, budget, status.

```
Select employee ID,
       First name,
       last name,
       department,
       salary,
       Project ID,
       Project name,
       budget,
       Status
FROM database schema. employees AS A
FULL OUTER JOIN database schema.employees
ON A.Employee ID = B.Employee ID
```

EmployeeID	First Name	Last Name	Department	Salary	Project ID	Project Name	Budget	Status
1	John	Doe	IT	70 000	101	AI development	100 000	Complete
2	Alice	Smith	HR	60 000	102	Employee training	50 000	Ongoing
3	Bob	Johnson	Finance	75 000	104	Financial Analyst	90 000	Ongoing
5	Emma	Wilson	Sales	65000	~~Null~~	~~Null~~	~~Null~~	~~Null~~
4	David	Brown	IT	72000	~~Null~~	~~Null~~	~~Null~~	~~Null~~
6	Michael	Clark	Finance	80 000	106	Risk Management	80000	Pending

7 Question

Retrieve employees who earn more than 70 000

- Return : Employee ID, First name, last name, department, salary

```
Select Employee ID,
       First Name,
       Last Name,
       Department,
       Salary,
WHERE  salary > 70 000
FROM database.schema.employees
```

Employees

Employee ID	First Name	Last Name	Department	Salary
3	Bob	Johnson	Finance	75000
4	David	Brown	IT	72 000
6	Michael	Clark	Finance	80000

8 Question

Retrieve employees ~~who~~ working in either IT or finance department

Return : Employee ID, First Name, Last Name, Department, Salary

Select Employee ID,
First Name,
Last Name,
Department,
Salary,
WHERE department = IT OR Finance
FROM database schema.employees

Employees

Employee ID	First Name	Last Name	Department	Salary
1	John	Doe	IT	70 000
3	Bob	Johnson	Finance	75 000
4	Dand	Brown	IT	72 000
6	Michael	Clark	Finance	80000

9 Question
Retrieve projects that are not yet completed
- Return : Project ID Project Name, Budget, status

Select Project ID
Project Name
Budget,
Status,
WHERE Status NOT 'complete'
FROM database.schema.projects

Projects

Project ID	Project Name	Budget	Status
102	Employee Training	50 000	ongoing
103	Cybersecurity Audit	75 000	pending
104	Financial Analysis	90 000	ongoing
106	Risk Management	80 000	pending

10 Question

Retrieve projects that have a budget greater than 70000 and are not completed

- Return project id, project name, budget, status

```
Select Project ID,
       Project Name,
       Budget,
       Status
WHERE budget > 70 000 AND status NOT 'Complete'
FROM database.schema.projects
```

Project ID	Project Name	Budget	Status
103	Cybersecurity Audit	75 000	pending
104	Financial Analysis	90 000	Ongoing
106	Risk Management	80 000	Pending

11 Question

Retrieve employees from New York OR Toronto, ordered by salary in descending order

- Return : Employee ID, First Name, Last Name, Department Salary, City

```
Select  Employee ID,
        First Name,
        Last Name,
        Department,
        Salary,
        City,
WHERE City = New York OR Toronto
ORDER BY Salary DESC
FROM database schema. employees
```

Employees

Employee ID	First Name	Last Name	Department	Salary	City
6	Michael	Clark	Finance	80 000	New York
3	Bob	Johnson	Finance	75000	Toronto
1	John	Doe	IT	70 000	New York

12 Question

Retrieve the top 3 highest-paid employees

- Return : Employee ID, First Name, Last Name, Departments, Salary

```
Select  Employee_ID,
        First_Name,
        Last_Name,
        Departments,
        Salary,
ORDER BY Salary DESC,
Limit 3;
FROM database.schema.employees
```

Employees

Employee ID	First Name	Last Name	Departments	Salary
6	Machael	Clark	Finance	80000
3	Bob	Johnson	Finance	75 000
4	David	Brown	IT	72 000

13 Question

Find the total salary per department sorted in descending order

- Return department, totalSalary (Rename SUM(Salary) as total Salary).

```
Select SUM(salary)
FROM database.schema.employees
GROUP BY department AS Total Salary
```

Employees

Department	Total Salary
IT	142 000
HR	60 000
Finance	155 000
Sales	65 000

14 Question

Find the average salary per city but only include cities where the average salary is greater than 65 000

```
Select City
      AVG(Salary) AS Average Salary
FROM database.schema.employees
GROUP BY City
HAVING AVG(Salary) > 65000
```

2 for each

Employees

City	Average Salary
New York	75 000
Toronto	75 000
London	72 000

15 Question

Count the number of employees per department, including only departments with more than 1 employee

- Return department, employee count (Rename Count (employee) as Employee Count).

```
Select  department,
        COUNT (EmployeesID) AS EmployeeCount
FROM database.schema.employees
GROUP BY department
HAVING  COUNT (EmployeeID) > 1
```

Employees

Department	Employee Count
IT	2
Finance	2

16 Question

Retrieve the number of projects per status, but only include statuses with at least 2 projects

- Return status, project count (Rename Count (Project ID) as ~~Select~~ Project Count).

```
Select  Status
        COUNT( Project ID) AS Project Count
FROM database.schema.project
GROUP BY Status
HAVING  COUNT (Project ID) ≥ 2
```

Project

Status	Project Count
Complete	2
Ongoing	2
Pending	2

~~14~~ Question 7

Retrieve the total project budget per employee, but only for employees who are managing projects more than 150 000

- Return Employee ID, First Name, Last Name, total Project budget (Rename sum(Budget) as Total Project budget).

```
Select  employee ID,
        First Name,
        Last Name,
FROM database.schema.employees AS A
INNER JOIN database.schema.projects AS B
ON A.Employee ID = B.Employee ID
~~GROUP BY WHERE~~
  GROUP BY Employee ID ~~AS Total Project Budget~~
  HAVING SUM(Budget) > 150000; AS Total Project Budget
```

Employee ID	First Name	Last Name	Total Budget Project
1	John	Doe	175 000